

Experiment 1-FTK Imager: A Forensic Imaging Tool Overview

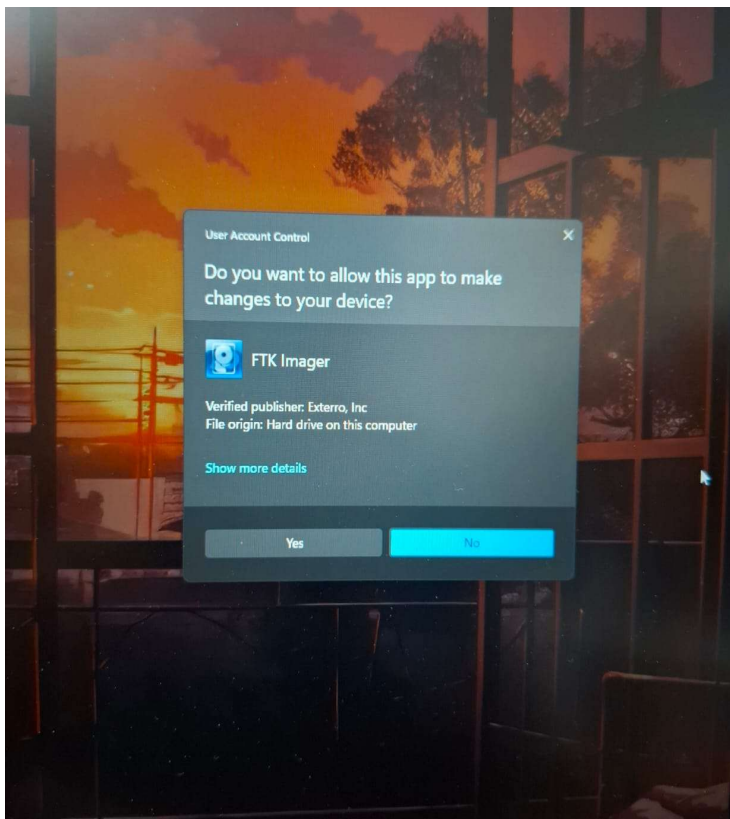
AIM:

Acquiring Volatile Memory (RAM) Using FTK Imager

Steps to Capture RAM Using FTK Imager

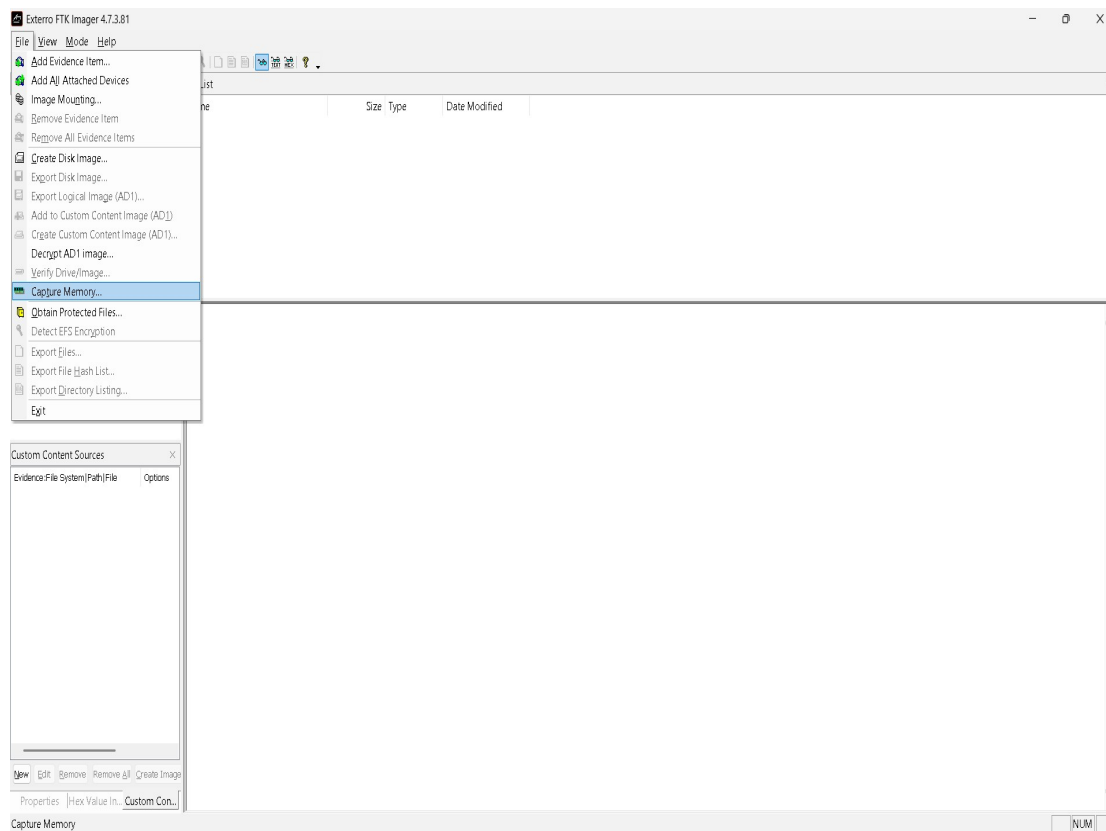
1. Run as Administrator

- Open **FTK Imager** with administrative privileges.
- Right-click the FTK Imager icon and select **Run as administrator**.



2. Initiate Memory Capture

- In the top menu bar, click **File** → **Capture Memory...** from the dropdown list



3. Configure Destination

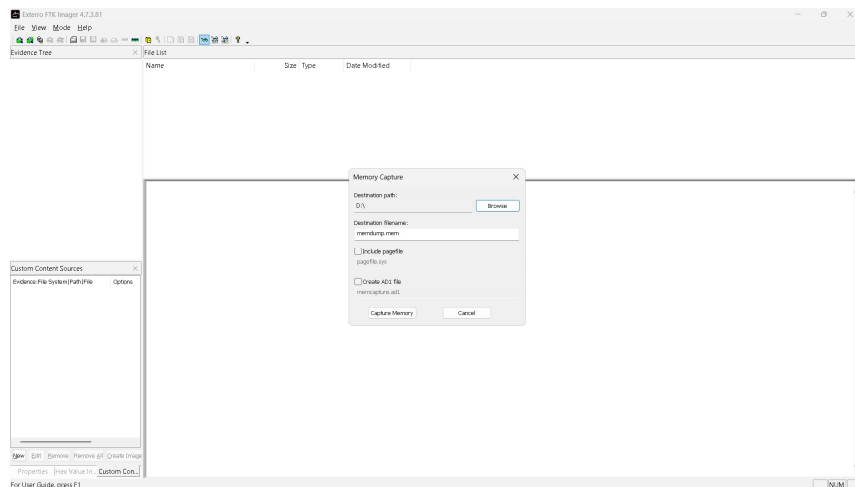
A dialog box will appear where you configure where and how the memory will be saved.

- **Destination Path:**
Click **Browse** to select a folder on an **external drive** (not the system drive).
- **Destination Filename:**
You can keep the default memdump.mem or assign a more descriptive name.
- **Optional: Include pagefile.sys**
Check this box to capture pagefile.sys, which is virtual memory stored on the disk.

Note: This may increase capture size and duration, but can contain valuable artifacts.

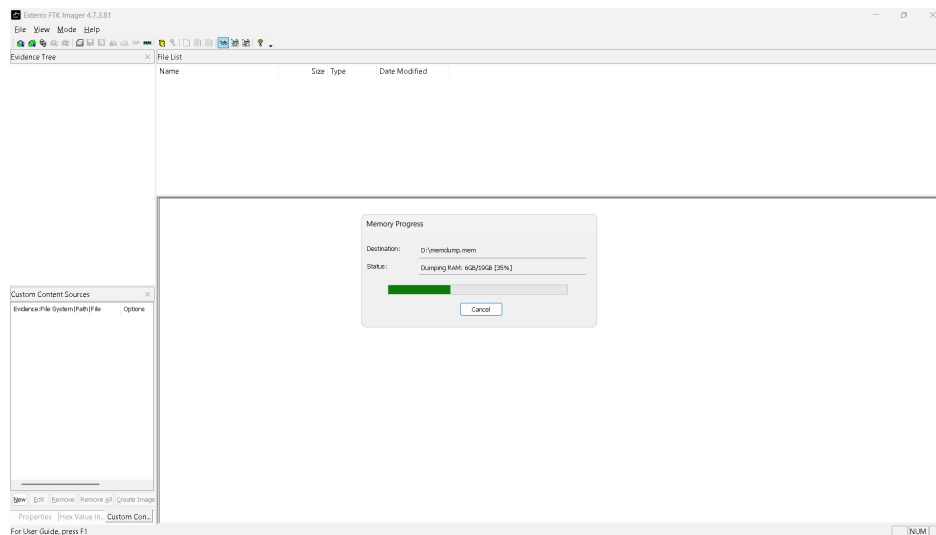
- **Optional: Create AD1 file**
Saves the memory dump in an AccessData-specific container file.

Generally not necessary if using tools like **Volatility** for analysis.



4. Start Capture

- Click the **Capture Memory** button to begin acquisition.



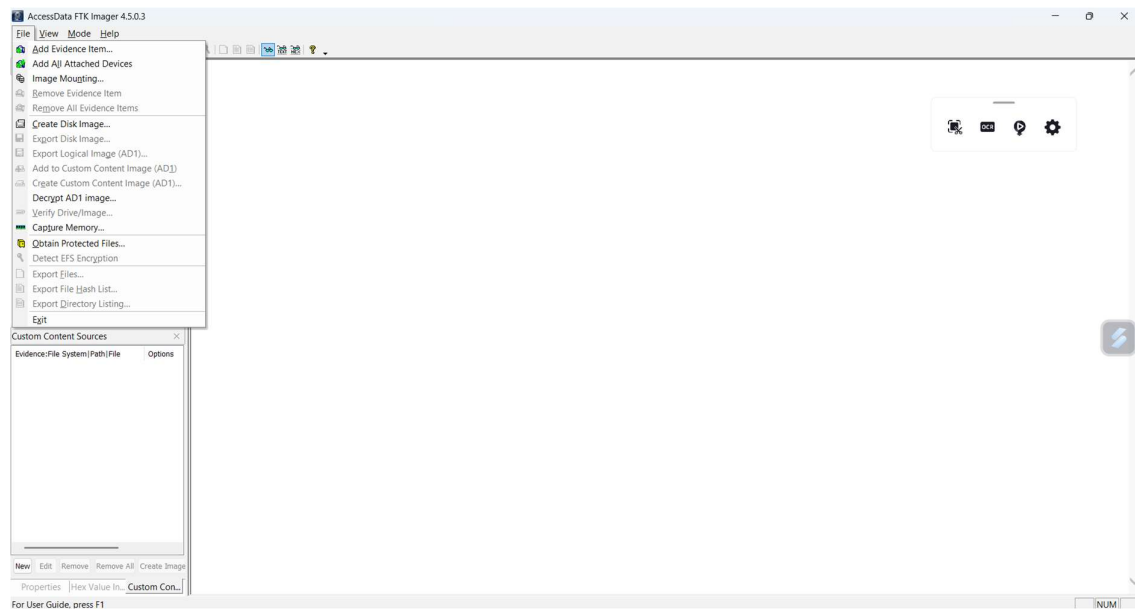
5. Wait for Completion

- A progress bar will indicate the capture status.
- Capture time depends on the system's RAM size.
- Once finished, the memory dump file will be available in the destination folder.

Acquiring Non-Volatile Memory (Disk Image) Using FTK Imager

1. Start the Process

- In FTK Imager, go to the top menu bar: **File → Create Disk Image....**

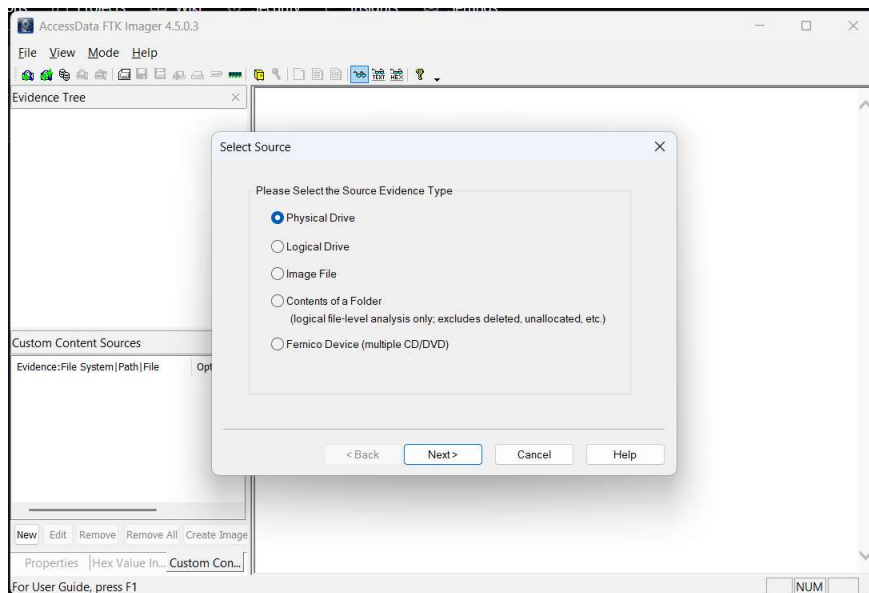


2. Select Source Evidence Type

A window will appear asking you to choose the source type:

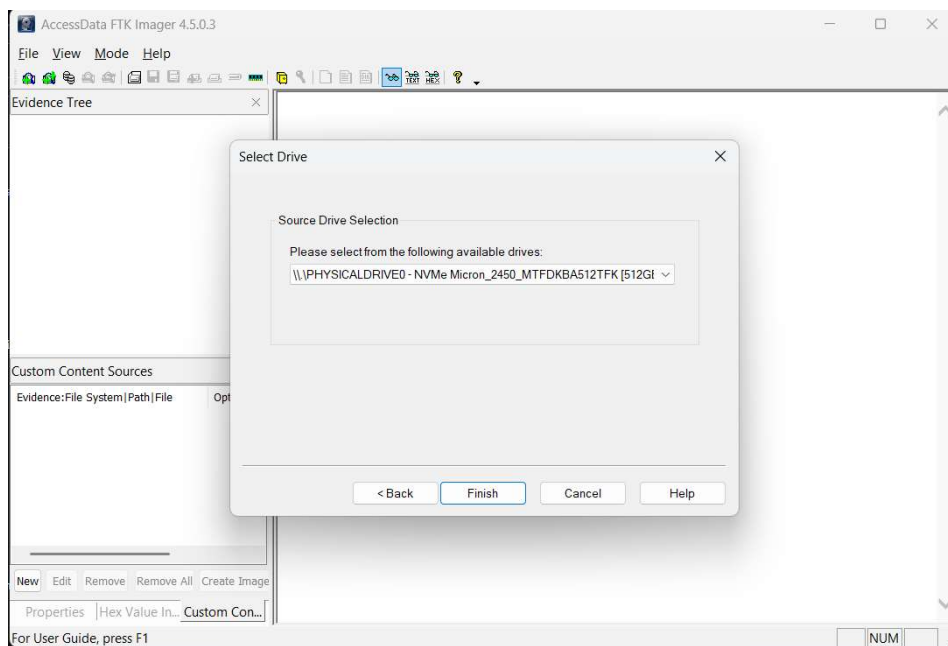
- **Physical Drive:** Images the entire disk, including all partitions, unallocated space, and the Master Boot Record (MBR).
- **Logical Drive:** Images a specific partition (e.g., C: drive).
- **Image File:** Converts or copies an existing image file.
- **Contents of a Folder:** Creates an image of a specific folder only.

Tip: For full forensic imaging, select **Physical Drive**.



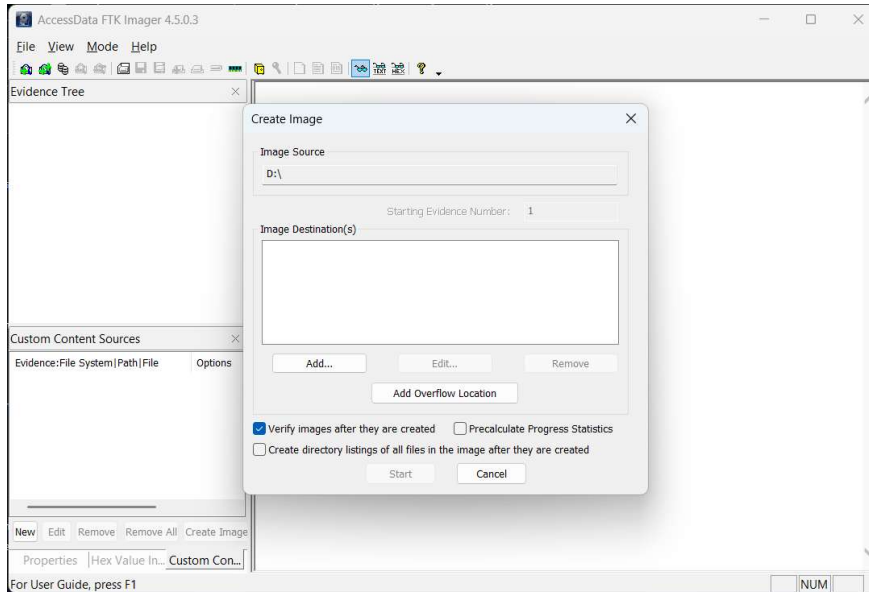
3. Select the Source Drive

- From the dropdown, choose the physical drive to image (connected via **write-blocker**).
- Click **Finish**.



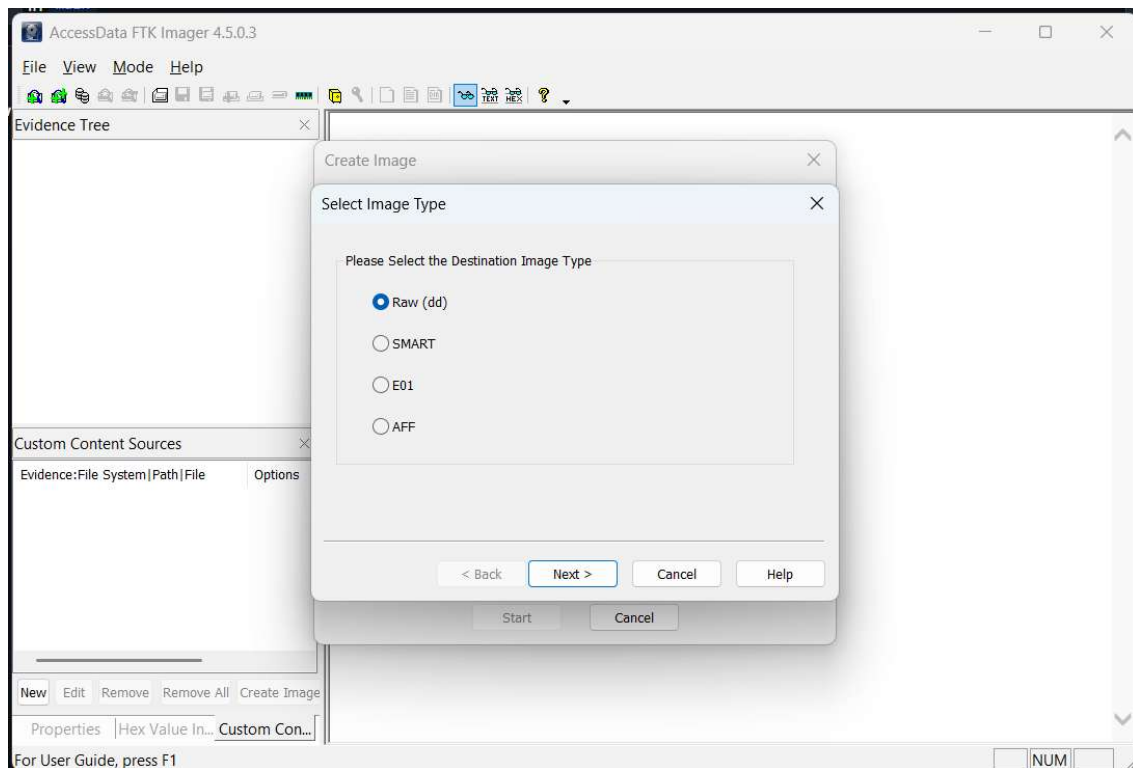
4. Configure the Image Destination

- Click **Add...** in the "Create Image" window to define the image **format** and **destination**.



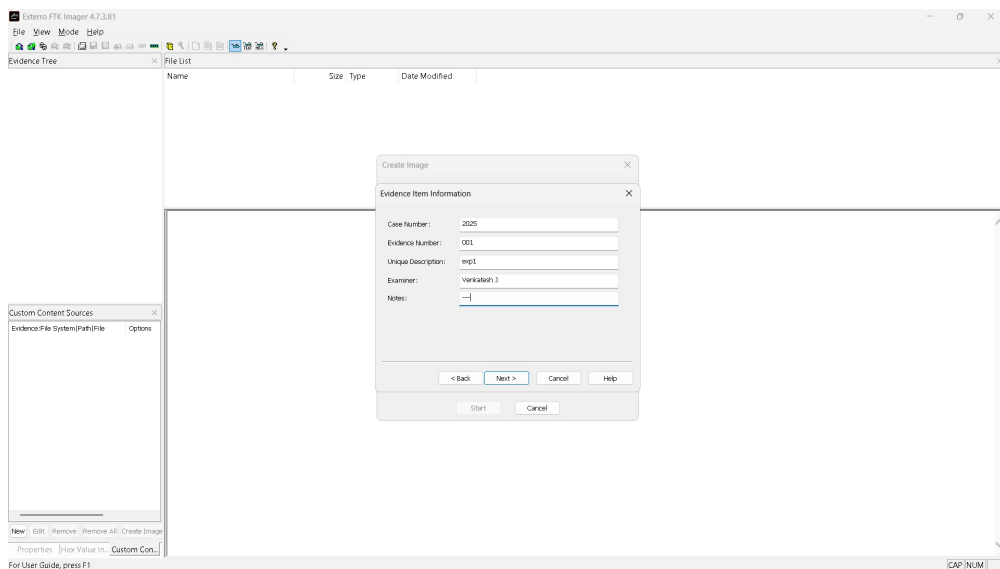
Options:

- **Image Type:**
 - **E01 (EnCase Format):** Recommended; includes compression, metadata, and error-checking.
 - **Raw (DD):** Bit-for-bit copy with no extra features.

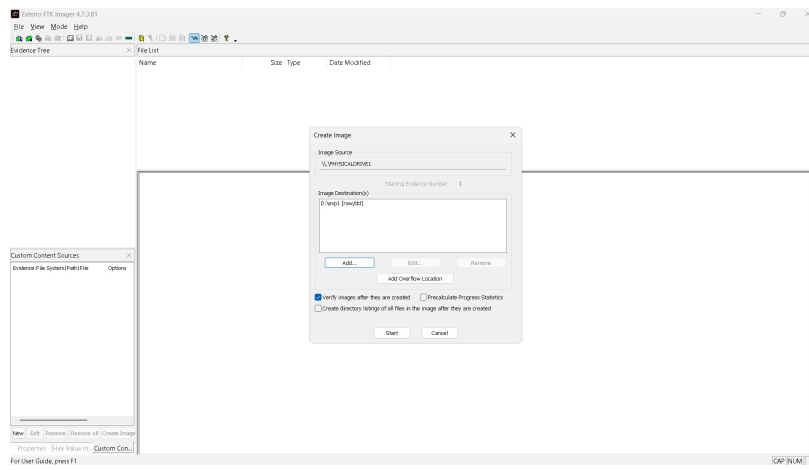


- **Fill in Evidence Item Information:**

- Enter case details, examiner name, and description.
- This information is stored in the image metadata (important for documentation).

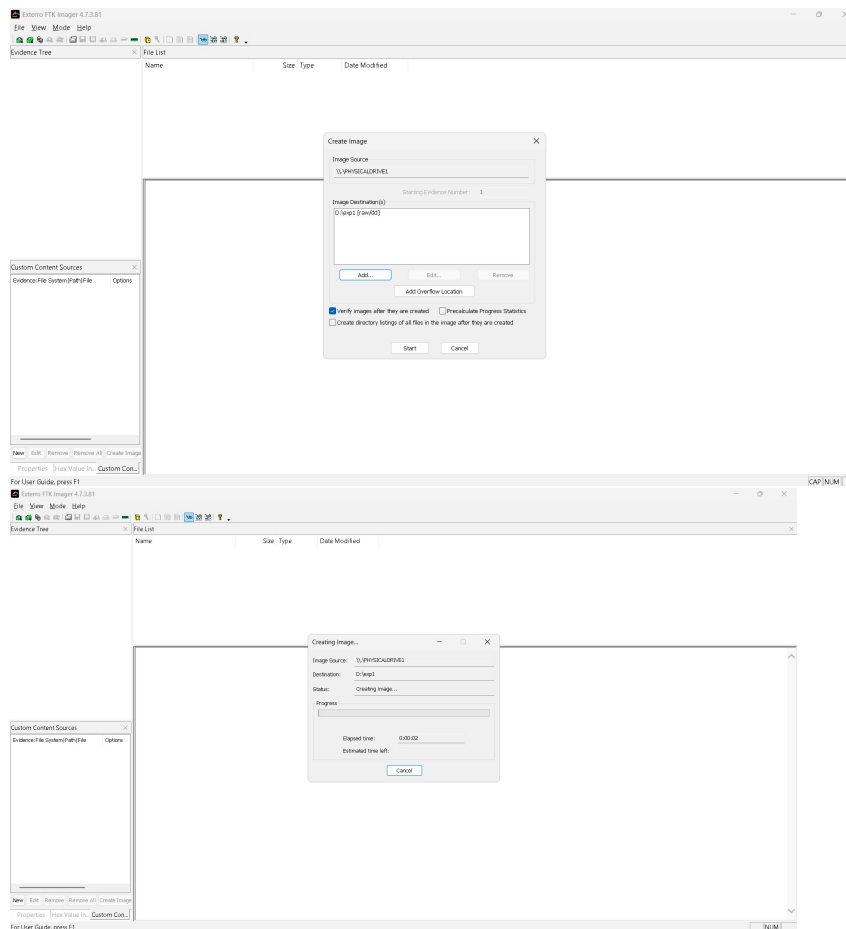


- **Choose Destination Folder:**
 - Must be a different drive from the source.
 - Name the image file (e.g., Case001_SuspectHDD).
- **Image Fragment Size:**
 - Set a value to split the image into multiple parts.
 - Set to 0 for a single image file.



5. Start the Imaging Process

- Click **Finish** to return to the "Create Image" screen.
- Check **"Verify images after they are created"** to calculate hash values and ensure integrity.
- Click **Start**.



6. Completion and Hash Verification

- The imaging process may take time depending on the drive size.
- After completion, FTK Imager verifies the hashes automatically.
- A final window shows **MD5** and **SHA1** hashes for both the source drive and image.
- Matching hashes confirm the forensic image's integrity.

Result:

Using **FTK Imager**, volatile memory (RAM) was successfully acquired from the target system. The tool created a complete forensic image of the system's memory in a secure format without altering the original data. The captured memory image was verified through hash values to ensure data integrity. This image can now be used for further forensic analysis, such as identifying running processes, open network connections, and potential malicious activities.

Conclusion:

The experiment demonstrates that **FTK Imager** is an effective and reliable tool for acquiring volatile memory in a forensic investigation. It ensures data integrity through hashing and provides investigators with crucial in-memory evidence. Thus, FTK Imager serves as a vital utility in digital forensics for capturing, preserving, and analyzing live system memory without compromising the original evidence.

Notes

- Always use a **write-blocker** to prevent modifications to the evidence.
 - **Hash verification** is critical to maintain a chain of custody and admissibility in court.
 - **Image Fragmentation** is useful for large drives or storage limitations.
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References

- [FTK Imager Official Website](#)
- FTK Imager Documentation